

Sana'a Factory, Yemen

PV-Storage-Diesel-Storage-type DVR Microgrid System Technical Proposal

Project Location: Bani Mattar District, Sana'a City, Yemen

Geographic Coordinates: 15.2811°N / 44.0811°E

Date: May 2026

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1. Project Overview

1.1 Project Background

This proposal designs a microgrid system for a clean workshop factory located in the Bani Mattar District of Sana'a City, Yemen (approximately 18 km from the capital). The system comprises a photovoltaic (PV) system, a battery energy storage system (BESS), a diesel generator (DG), and a storage-type dynamic voltage restorer (DVR). The system aims to achieve uninterrupted power supply during 10 working hours while optimizing the cost-performance ratio of PV and storage, meeting the demand of loads highly sensitive to power quality.

1.2 Load Demand

NO.	Load Type	Power (kW)	Remarks
1	Air conditioning load	200	Constant operation
2	Air compressor	200	Variable frequency start
3	Automated filling line	~300	Includes purification, injection, blow molding, filling, labeling
4	PLCs and servo motors, etc.	-	Highly sensitive to power quality
Total	Average total load	650~700	Sequential equipment start-up required

The plant plans to install a 1000 kVA transformer and configure an 800 kW containerized diesel generator (or reserved engine room). Based on 10 hours/day and 700 kW total load, daily average electricity consumption ≥ 7000 kWh.

1.3 Design Requirements

- Uninterrupted power supply during 10 working hours
- Optimal cost-performance ratio between PV and storage
- Voltage fluctuation $\leq \pm 5\%$, meeting sensitive load power quality requirements
- When PV or storage fails, diesel generator responds immediately with transfer time ≤ 5 ms

- Storage DVR system supports up to 20 minutes of emergency power
- PCS must have grid-forming capability
- All power electronic devices shall be integrated into the plant SCADA system via Modbus TCP/IP

1.4 Site Conditions

Parameter	Value	Remarks
Latitude	15.2811°N	Low-latitude tropical highland climate
Longitude	44.0811°E	
Time zone	UTC+3	Early sunrise, late sunset, very long sunshine duration
System efficiency	82%	Comprehensive PR value
Peak generation period	8:00-16:00	Local peak sunshine hours
N-type module degradation (10y)	≤8%	Optimized LCOE
SOH Battery SOH after 10 years	≥85%	LFP chemistry
Spring (Mar–May)	Strongest sunlight, mild temperatures	Peak annual generation
Summer (Jun–Sep)	Occasional clouds, dust	Slight reduction in generation
Autumn/Winter (Oct–Feb)	Mostly sunny, stable	Steady generation

2. System Architecture

2.1 System Composition

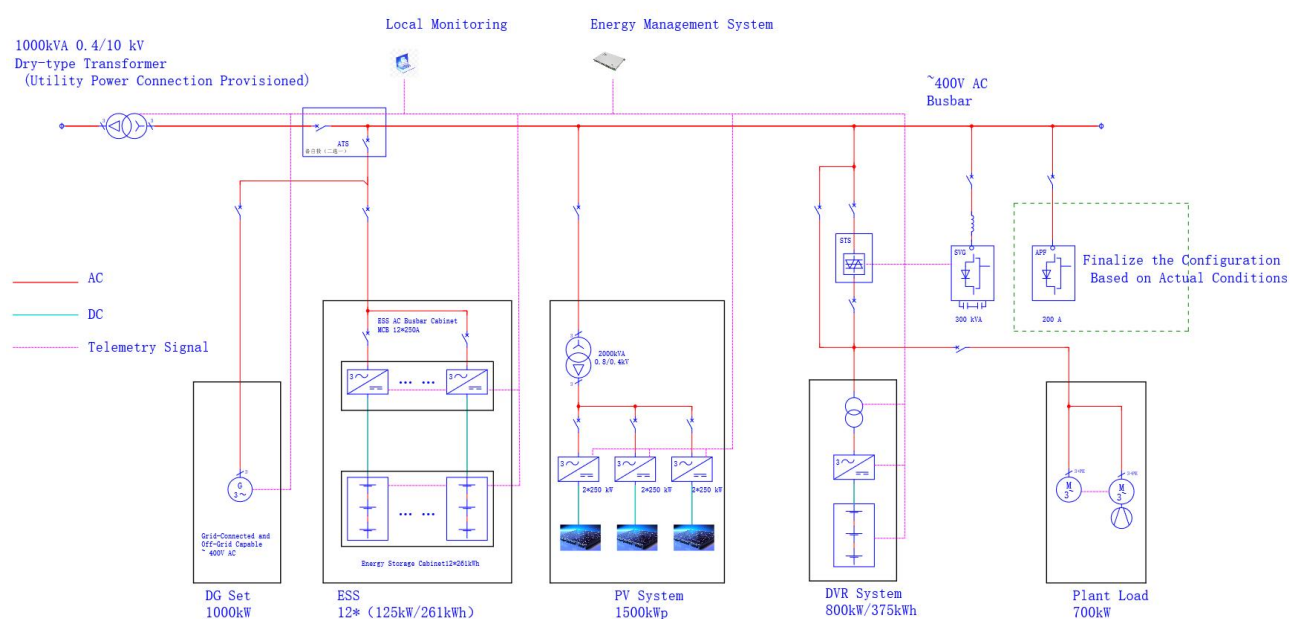
This microgrid consists of four core subsystems connected in parallel to a 400 V low-voltage busbar, coordinated by an Energy Management System (EMS). The architecture is as follows:

- PV system: ≥ 1500 kWp, N-type monocrystalline silicon modules, connected via three 400 V feeders.
- BESS: ≥ 3132 kWh, twelve 261 kWh outdoor battery cabinets + two PCS cabinets of 750 kW total, grid-forming operation.
- Diesel generator: 1000 kW containerized, auto start/stop, for emergency backup and peak shaving.
- Storage-type DVR: 800 kW / 375 kWh, supports up to 20 minutes, transfer time ≤ 5 ms, ensures power quality.

2.2 Primary System Topology

The system uses a 400 V low-voltage common busbar where all sources and loads are connected in parallel.

PV/ESS/DG Hybrid Microgrid System



- PV system connects via grid-following inverters to the 400 V bus, three independent MPPT controllers.
- BESS connects via grid-forming PCS, with black-start and islanding capability.
- DG connects via automatic transfer switch (ATS) and can synchronize for parallel operation.
- Storage-type DVR is connected in series between the bus and sensitive loads, providing real-time voltage support.
- 1000 kVA transformer serves as utility interface (if available) or reserved grid connection point.

2.3 System Layout

The PV array is installed on factory rooftops and open areas, using fixed-tilt racks (optimal tilt $\sim 15^\circ$). Battery cabinets are placed in a dedicated outdoor area with fire access. The DG is containerized or placed in a reserved room. DVR and PCS equipment are located inside the electrical room.

3. Photovoltaic System Design

3.1 Installed Capacity

Total PV capacity ≥ 1500 kWp, using N-type TOPCon or HJT high-efficiency monocrystalline modules with efficiency $\geq 22.5\%$, first-year degradation $\leq 1\%$, linear degradation $\leq 0.4\%$ per year, resulting in $\sim 8\%$ power reduction after 10 years.

3.2 Module Selection

Parameter	Specification	Remarks
Module type	N-type TOPCon monocrystalline	High efficiency, low degradation
Single module power	580Wp	Mainstream large-format
Module efficiency	$\geq 22.5\%$	Industry leading
First-year degradation	$\leq 1\%$	
Annual degradation	$\leq 0.4\%$	Linear
Efficiency after 10 years	$\geq 92\%$	Better than average
Estimated quantity	≥ 2586 pieces	1500kWp / 580Wp
Estimated total area	$\geq 8,500\text{m}^2$	Approx. 3.3m^2 per module

3.3 Inverter Configuration

String grid-following inverters, three feeders to 400 V bus, each ~ 500 kWp. Inverter efficiency $\geq 98.6\%$, support active/reactive power regulation, low-voltage ride-through (LVRT) capability.

3.4 Mounting and Installation

- Mounting: Fixed-tilt rack, optimal tilt $\sim 15^\circ$ (close to local latitude)
- Orientation: True south to maximize annual yield
- Row spacing: No shading from 9:00 to 15:00 on winter solstice
- Corrosion protection: Class C4 (for tropical coastal environments)

3.5 Energy Yield Estimation

Month	Daily average generation (kWh)	Remarks
Jan, Feb, Mar, Apr, Oct, Nov, Dec	≥9000	Peak generation, strongest sunlight
May, Sep	≥7500	Occasional clouds/dust, slight reduction
Jun, Jul, Aug	≥6200	Mostly sunny, stable generation
Annual daily average	≥7500	System efficiency 82%
Annual generation	≥27,375,00	Approx. 1825 kWh/kWp/year

4. Battery Energy Storage System Design

4.1 Installed Capacity

Total BESS capacity ≥ 3132 kWh, using twelve 261 kWh outdoor cabinets. Each cabinet integrates LFP battery cluster, grid-forming PCS, BMS, thermal management, and fire suppression.

4.2 Battery Selection

Parameter	Specification	Remarks
Cell chemistry	LFP (Lithium Iron Phosphate)	High safety, long life
Single cabinet capacity	261kWh	Standardized outdoor cabinet
Number of cabinets	12 套	Total capacity 3132 kWh
System total capacity	≥ 3132 kWh	Meets 10-hour operation requirement
Cycle life	≥ 6000 cycles	@80% DoD
SOH after 10 years	$\geq 85\%$	Meets degradation requirement
Round-trip efficiency	$\geq 85\%$	RTE
Operating temperature	$-20^{\circ}\text{C} \sim +55^{\circ}\text{C}$	Suitable for Yemen's tropical climate

4.3 PCS Configuration

Twelve PCS units, each ≥ 125 kW, meeting full load charge/discharge requirements. PCS capabilities:

- Grid-forming: Independently establish 400 V/50 Hz microgrid voltage and frequency reference.
- Black-start: Autonomous recovery after full system outage.

- Independent active/reactive power control: Participate in microgrid frequency and voltage regulation.

4.4 Safety and Fire Protection

- Each cabinet: Pack-level fire suppression (Novec 1230 or aerosol)
- BMS three-level protection: cell, module, system.
- Outdoor cabinet: IP55, corrosion class C4.
- Thermal management: Hybrid liquid cooling + air cooling, suitable for ambient temperatures above 45°C.

5. Diesel Generator System

5.1 Installation Scheme

One 1000 kW containerized diesel generator (or reserved room installation) as the ultimate backup emergency power source.

1. Equipment start-up characteristics:

- Air conditioning: Variable frequency, start current 1.5–2 rated, peak ~300 kW for ~200 s.
- Air compressor: Variable frequency, start current 1.5–2 rated, peak ~300 kW.
- Filling line: Multiple servo motors, variable frequency start, start current ~3 rated, peak ~600 kW.
- Worst-case total start peak: $300+300+600 = 1200$ kW.

2. Power redundancy requirement:

To keep voltage fluctuation $\leq \pm 5\%$ for sensitive loads, sufficient power margin is needed.

Calculated required generator power $\approx$ (Total load power + Maximum single motor power $\times$ Starting multiplier) $\times$ 1.2

With sequential start-up, the largest single motor ≤ 75 kW with start multiplier ≤ 3 , total DG power can be kept within 1000 kW.

Tentative DG rating: 1000 kW (PRP) or 1250 kVA, containerized silent type.

- Automatically synchronize and parallel with storage PCS when SOC is low or PV output is persistently low.
- Independent power source during storage maintenance.
- Emergency backup (DVR supports for 5 ms while DG starts and transfers).

5.2 Technical Parameters

Parameter	Specification	Remarks
Rated power	1000kW	Prime power (PRP)
Rated voltage	400V	Three-phase four-wire
Frequency	50Hz	
Start mode	Auto/manual	Receives start command from EMS

Parameter	Specification	Remarks
Start response time	≤3min	From start signal to full load
Fuel type	Diesel	Locally available in Yemen
Installation	Containerized / room	Noise reduction, dust protection
Paralleling capability	Yes	Can parallel with storage

5.3 Operation Strategy

- Emergency: DG starts immediately upon PV or storage failure.
- Peak shaving/backup: DG auto-starts and parallels when storage SOC ≤ set threshold (e.g., 20%)
- SOC recovery: DG runs to charge storage until SOC recovers to set value (e.g., 80%), then DG exits.
- Night operation: Lighting powered by remaining storage; if insufficient, DG supplements.
- Maintenance mode: DG supplies power alone when storage is manually disconnected

6. Storage-type DVR System Design

6.1 System Positioning

The storage-type DVR is key to ensuring power quality. Sensitive loads (PLCs, servo motors, automated lines) require voltage fluctuation $\leq \pm 5\%$. The DVR is connected in series between the bus and sensitive loads, detecting and compensating voltage sags, swells, harmonics, etc., in real time.

6.2 Technical Specifications

Parameter	Specification	Remarks
Rated power	800kW	Covers all loads
Storage support time	20 minutes	LFP battery
Transfer time	$\leq 5\text{ms}$	Meets sensitive load requirements
Voltage compensation range	0% ~ +120%	Deep voltage conditioning
Compensation type	Sag, swell, flicker, harmonics	LFP battery
Response time	$\leq 5\text{ms}$	From detection to compensation
Topology	Series + storage	Online operation
Efficiency	$\geq 97\%$	Very low losses in normal operation

6.3 Working Principle

The DVR injects compensating voltage via a series transformer to keep load voltage constant. In normal conditions, DVR is in standby (zero power draw). When bus voltage exceeds the threshold, within 5 ms the DVR discharges storage and injects compensation voltage, keeping load voltage fluctuation within $\pm 5\%$.

6.4 Operation Scenarios

- Scenario 1 – DG transition: DVR maintains power quality during DG start-up and stabilization.

- Scenario 2 – PV/storage fault:DVR supports ~20 minutes, buying time for DG start and transfer.
- Scenario 3 – Microgrid voltage disturbance:PCS built-in DVR function switches to voltage source within 5 ms, forcing support while triggering DG start.

7. Energy Management System (EMS)

7.1 System Architecture

The EMS is the "brain" of the microgrid, coordinating PV, storage, DG, and DVR – all AC-coupled so that failure of any one subsystem does not affect the others. It achieves optimal energy dispatch with a layered architecture:

- Scheduling layer: Global energy optimization schedule.
- Control layer: Microgrid central controller (MGCC) for second-level real-time control, mode switching, frequency/voltage regulation.
- Device layer: Local controllers execute commands and report status.

7.2 Core Functions

- PV power detection and decision-making.
- Storage charge/discharge scheduling for lifetime extension and cost reduction.
- DG intelligent start/stop based on SOC, PV output, load demand.
- DVR coordination to pre-reserve energy for power quality events.
- Black-start management for orderly restoration after full outage.

7.3 Communication Integration

Interface	Protocol	Connected Systems
Modbus TCP/IP	Ethernet	PLC/SCADA
Modbus TCP/IP	Ethernet	EMS
Modbus TCP/IP	Ethernet	BMS
Modbus TCP/IP	Ethernet	PCS, DG, SVG, etc.
IEC 61850	Ethernet	Protection devices (optional)
OPC UA	Ethernet	Upper-level monitoring

8. System Operation Strategies

8.1 Five Weather Condition Strategies and Fault Modes

Condition 1 – Sunny, high irradiance (PV at full output)

PV output ≥ 700 kW, fully meeting load. Excess PV charges storage, SOC rises. If storage is full and PV still has surplus, MPPT curtailment reduces PV output.

Condition 2 – Cloudy, intermittent (PV fluctuating)

PV output 300–600 kW. Storage fills the gap; SVG compensates short-term fluctuations. EMS dynamically adjusts storage power to keep bus stable.

Condition 3 – Overcast/rainy (low PV output)

PV output ≤ 200 kW, mainly storage power. When SOC drops to threshold (e.g., 30%), EMS auto-starts DG to parallel, DG takes main load and charges storage. DG exits when SOC reaches 80%.

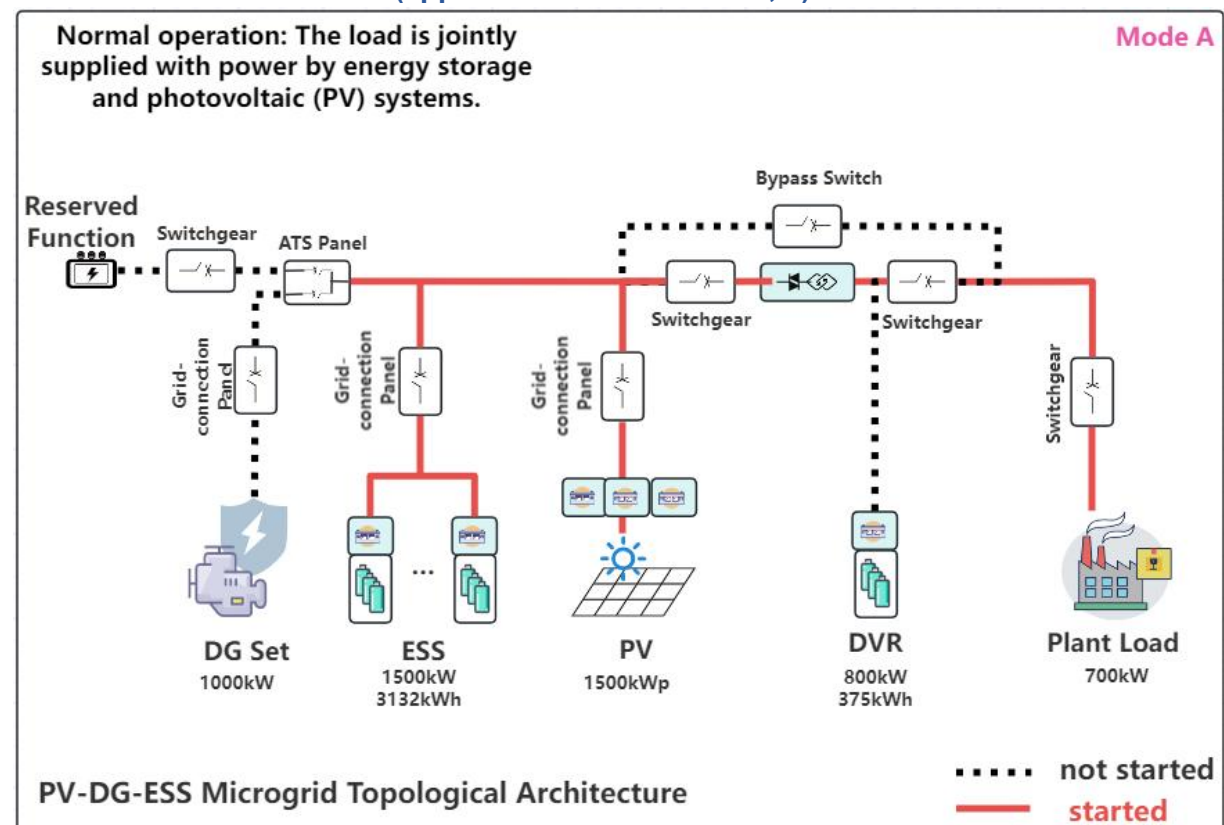
Condition 4 – Sandstorm/extreme weather (PV nearly zero)

PV negligible, storage alone. When SOC reaches 20%, DG starts. If SOC falls below 5%, storage DVR intervenes (up to 20 min) until DG stabilizes, then transfers to DG.

Condition 5 – PV or storage system fault

Regardless, if both PV and storage fail, DG responds immediately (≤ 5 ms), DVR supports (up to 20 min) until DG stabilizes, then transfers to DG emergency supply.

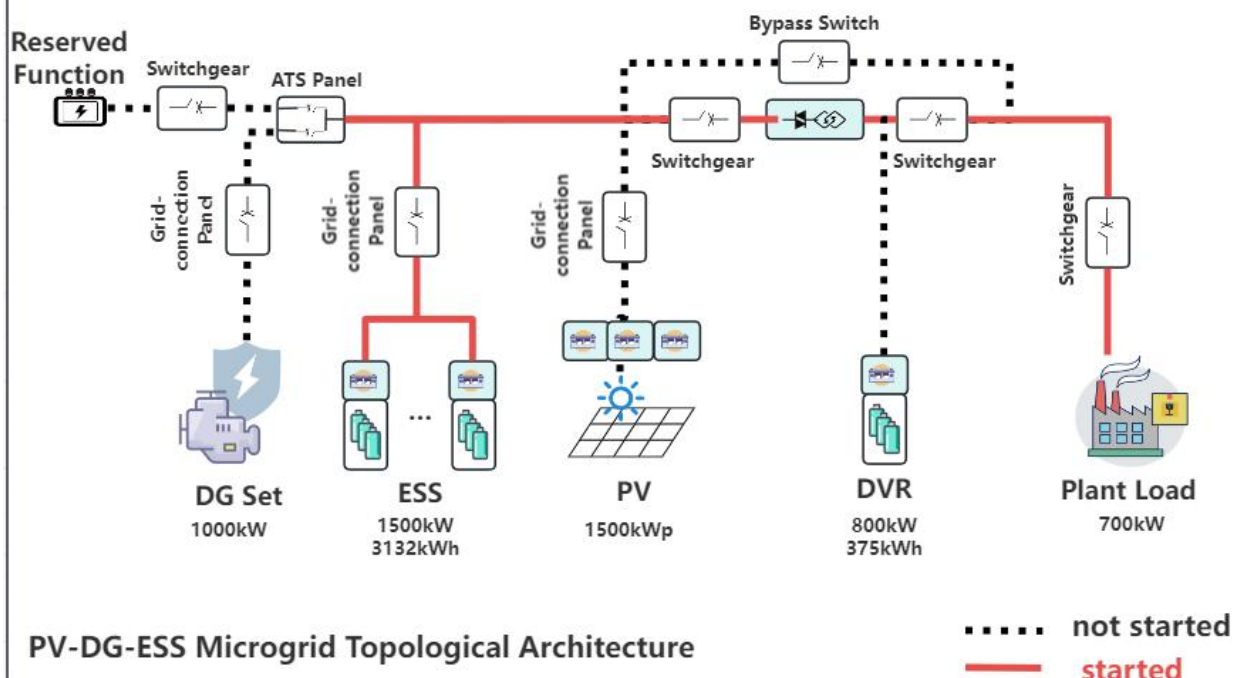
Mode A – Normal state (applicable to Conditions 1, 2)



Mode B – Storage normal, PV fault (applicable to Condition 3)

Adverse weather or PV system fault (ESS normal):
ESS powers the load solely. (State of Charge: $80\% \geq SOC \geq 20\%$)

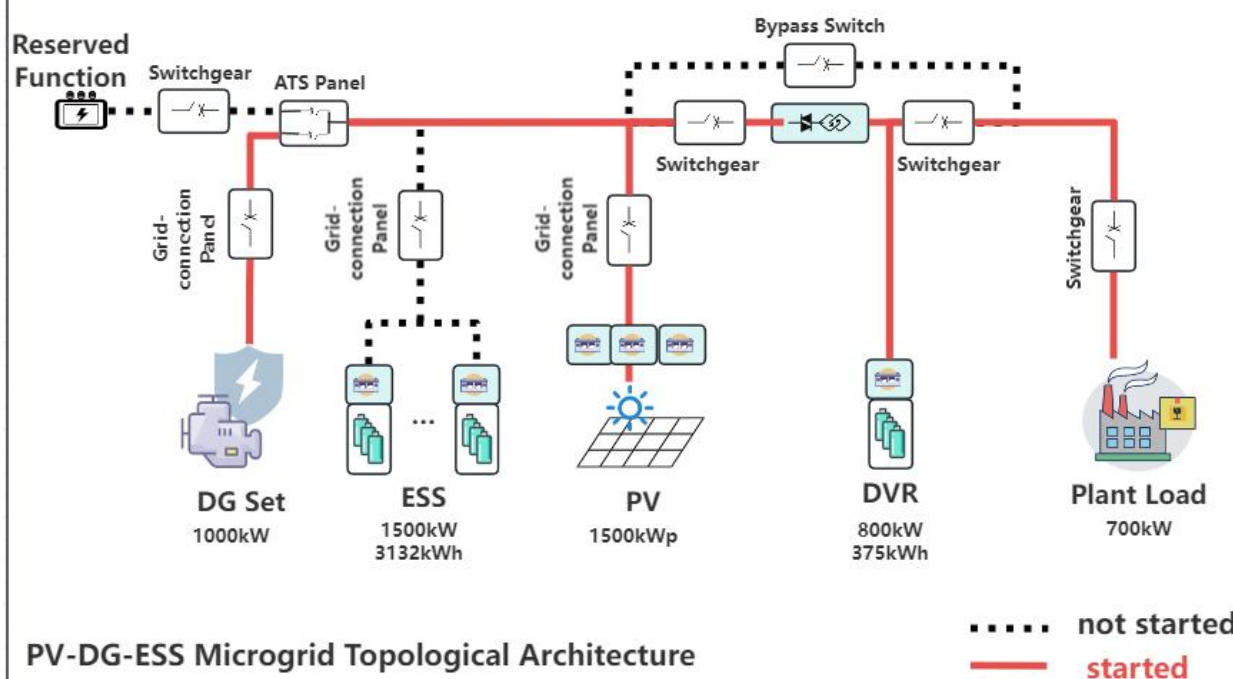
Mode B



Mode C – PV normal, storage fault (applicable to Conditions 4, 5 transition)

ESS fault, voltage fluctuation $\geq 5\%$ or ESS SOC $\leq 5\%$: DVR activates instantly and the DG set starts to supply power. Normal PV system can operate jointly with the DG set.

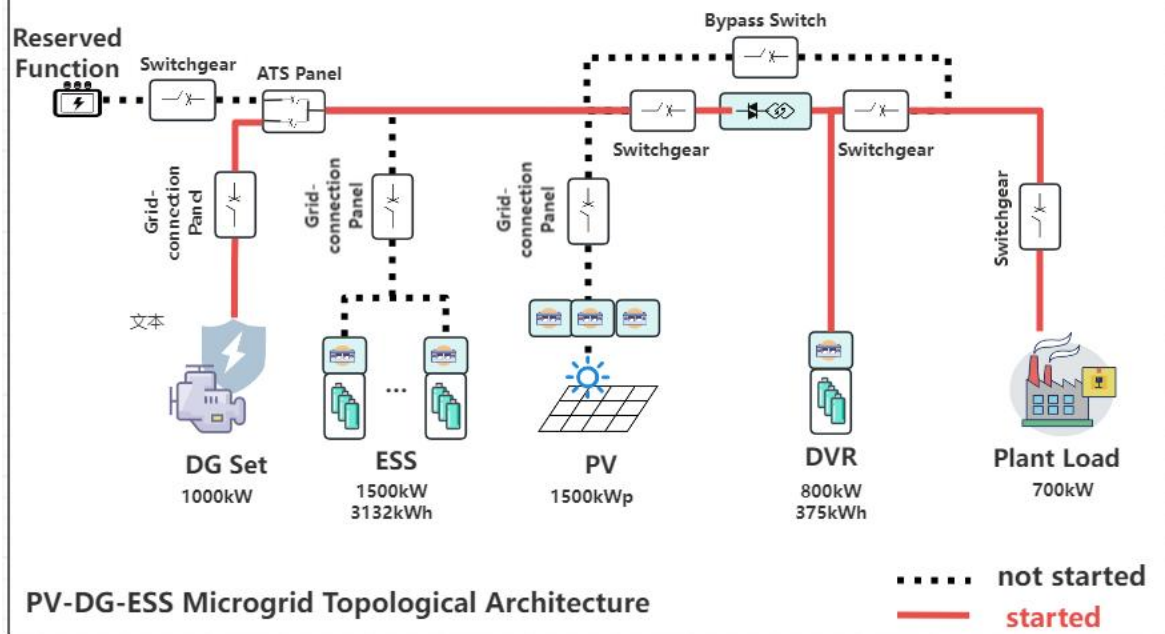
Mode C



Mode D – Both PV and storage fault (during Conditions 4,5)

ESS and PV simultaneous fault, voltage fluctuation $\geq 5\%$ or ESS SOC $\leq 5\%$:
DVR activates instantly and the DG set starts to supply power.

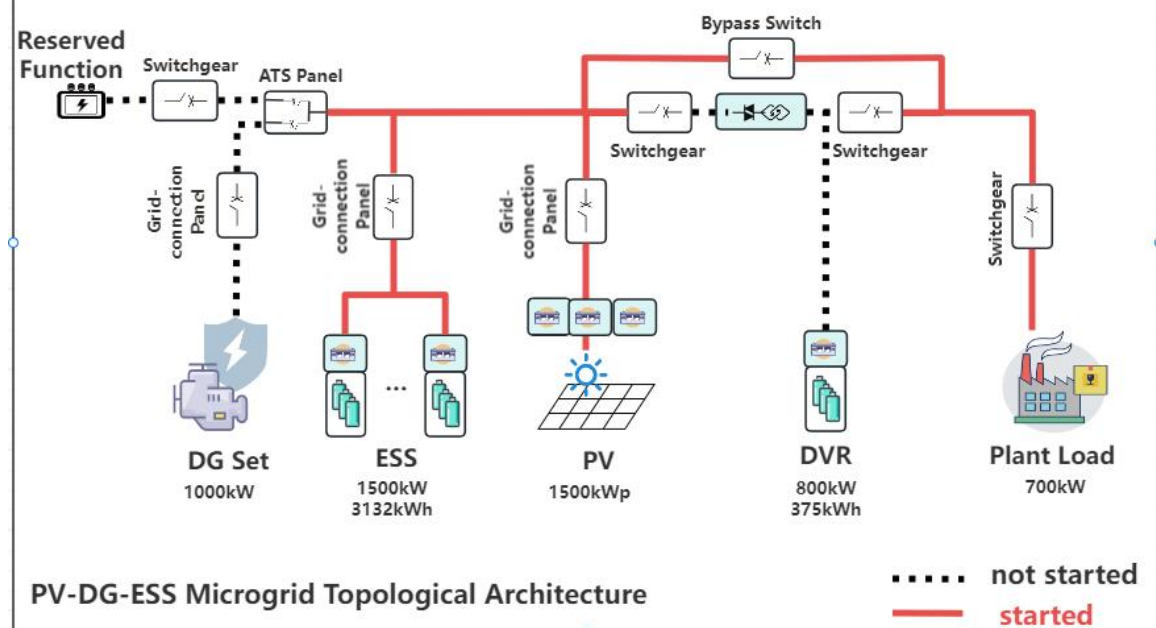
Mode D



Mode E – DVR fault

DVR System fault: Close bypass switch, open D2 and D3 switches.

Mode E



8.2 Morning Start-up Strategy

At 8:00 when production starts, storage must have reserved energy (suggested

SOC $\geq 30\%$) to meet startup inrush currents for the 200 kW air compressor and filling line. DG is backup; if reserved SOC insufficient, DG auto-starts to assist.

By 7:45 daily, EMS checks if SOC $\geq 20\%$. If not, start DG 15 minutes in advance to parallel; load is shared by DG + storage. Air compressor VFD soft-start, air conditioners PLC-staged, production line step-by-step start, keeping peak current below 1200 A (instantaneously supported by storage PCS). After start-up, if PV is normal, gradually ramp down DG.

8.3 Energy Storage Manual Maintenance (Special Operating Conditions)

When EMS determines severe PV fault condition (SOC $< 20\%$):

DG starts in parallel, charges until SOC $\geq 80\%$ or PV fault clears, then PV auto-synchronizes, DG exits without shock.

When EMS determines severe PV+storage fault:

System automatically transfers all active power to DG. DG immediately starts in island mode, DVR connects; after DG voltage stabilizes, DVR disconnects, loads fully supplied by DG. Manual disconnection and EMS selects storage isolation mode both AC and DC switches open for safe isolation. Storage cabinets can be serviced. After maintenance, manual closing of switches, exit isolation mode, EMS decides whether to exit DG. If DG exits, DVR connects, storage island mode supplies power, voltage stabilizes, PV connects, DVR disconnects, then storage+PV supply.

8.4 Diesel Generator Intervention/Exit Logic

Trigger Condition	Action	Threshold
Storage SOC below lower limit	DG auto-start	SOC $\leq 20\%$ (adjustable)
PV+Storage $<$ load for 5 min	DG auto-start	Persistent deficit
PV/storage fault	DG emergency start	Immediate
Storage SOC recovers to upper limit	DG exit	SOC $\geq 80\%$ (adjustable)
Load drops to night level	DG exit	Storage alone sufficient
Storage maintenance mode	DG alone supply	Manual trigger

8.5 Night Operation

At night (17:00 – 8:00), factory load is mainly lighting, small. Priority is storage supply. If storage SOC insufficient, DG starts briefly to supplement. During this period, storage can receive trickle charge for maintenance.

9. Main Equipment List

9.1 PV System

No.	Equipment	Specification	Qty
1	PV module	580 Wp, efficiency $\geq 22.5\%$	$\geq 2,580$ pcs
2	String inverter	150 kW, efficiency $\geq 98.6\%$	10units
3	Fixed mounting system	15° tilt, C4 corrosion	1lot
4	AC combiner panel	800V, 2500A	1
5	AC grid-connection panel	400V, 2500A	1
6	Lightning & grounding	according to IEC62305	1set

9.2 Energy Storage System

NO.	Equipment	Specification	Qty
1	Outdoor battery cabinet	261 kWh, LFP, liquid cooling, BMS, fire suppression	12
2	PCSGrid-forming PCS	125kW, AC400V	12
3	Storage AC combiner panel	12* 250A 400VAC	1
4	EMS for storage	The control system meets the dual requirements of communicating with the power dispatching platform (upper level) and accessing field devices (lower level), is compatible with Modbus TCP/RTU protocols, and enables comprehensive data acquisition, measurement, and control of the energy storage system. It features multi-energy coordinated control, grid-forming mode, anti-reverse power flow/demand	1set

NO.	Equipment	Specification	Qty
		management, and black start capability. The system supports complex local strategies and ensures millisecond-level command delivery latency.	

9.3 Diesel Generator and Distribution System

NO.	Equipment	Specification	Qty
1	Diesel generator set	1000kW, 400V/50Hz	1set
2	ATS panel	400V 2500A, 4P	1
3	DG synchronizing panel	Including grid-connected contactor, synchronizing relay, and multi-function meter.	1
4	Container / enclosure	Noise-reducing, dust-proof	1set

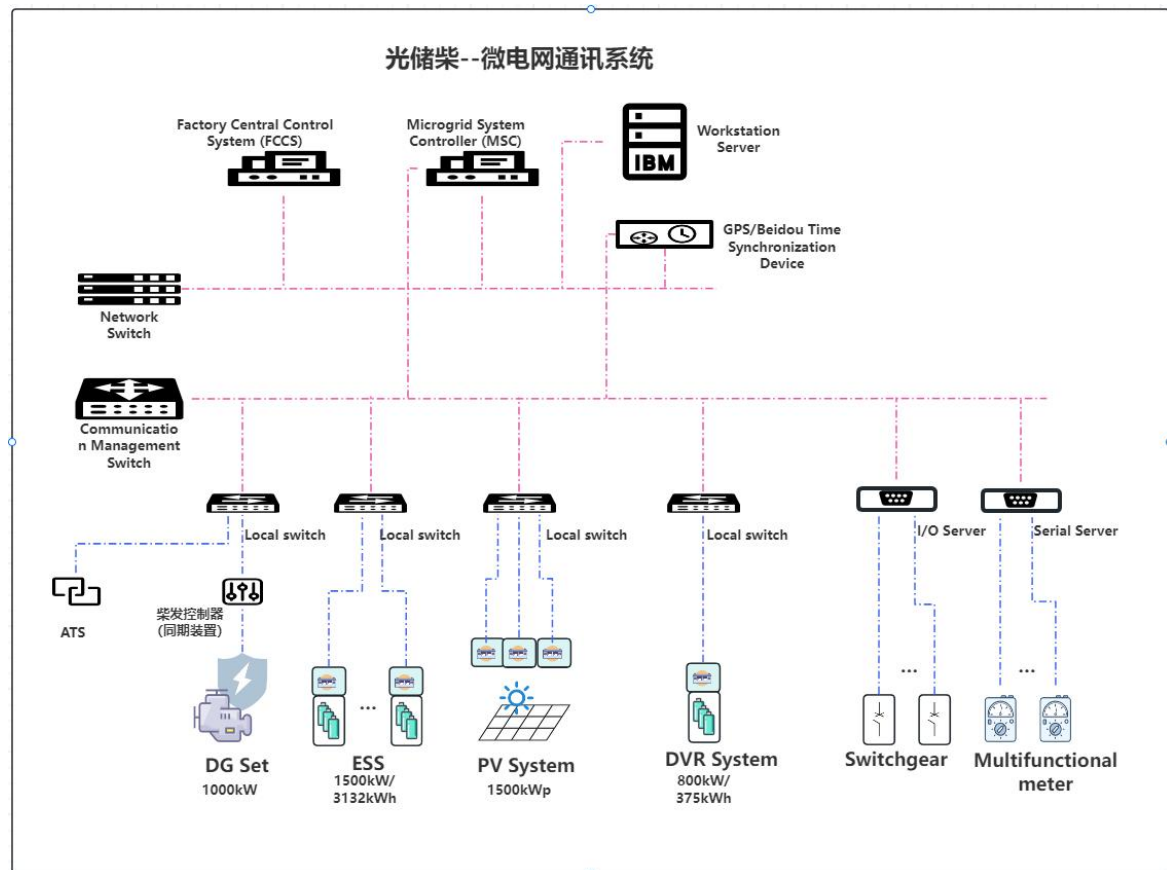
9.4 Storage DVR and Distribution System

NO.	Equipment	Specification	Qty
1	DVR cabinet	800 kW, 375 kWh, 163 Ah, 20 min support	1
2	Isolation transformer cabinet	800 kVA matched to DVR	1
3	Static transfer switch (STS) cabinet	2000kW	1
4	Battery cabinet	1P240S,3.2V,163Ah	3
5	Bypass switchgear	ACB 2500A -400V /AC	1
6	Dry-type transformer	1000kVA 10/0.4 kV	1
7	SVG cabinet	300kVA-400V/AC	1

NO.	Equipment	Specification	Qty
8	LV incoming (utility)	ACB 2000A -400V /AC	1
9	LV parallel incoming (storage)	ACB 2500A -400V /AC	1
10	LV parallel incoming (PV)	ACB 2500A -400V /AC	1
11	LV outgoing (loads)	ACB 2000A -400V /AC	1
12	Container (DVR+control)	Noise-reducing, dust-proof	1set
13	Container (distribution)	Noise-reducing, dust-proof	1set

9.5 Energy Coordination Management System

NO.	Equipment	Specification	Qty
1	Microgrid coordination control panel	Microgrid coordinator, smart gateway, GPS/Beidou clock, industrial Ethernet switch, data server, operator workstation, O&M software, system cabinet	1
2	UPS (Uninterruptible Power Supply)	Online type, 6kVA, 220VAC, including battery pack (backup time ≥ 2 hours); used for servers, workstations, switches, etc.	1



10. Economic Analysis and Investment Estimate

10.1 Investment Estimate

System	Main Investment Items	Estimated Amount (USD 10k)	Remarks
PV System		59.7	
Equipment Details	PV modules	42.6	2580pcs
	Mounting structure	9.95	1lot
	WInverters 150 kW	4.15	10units
	AC grid panel 400V/2500A	1.5	1
	AC combiner panel 400V/2500A	1.5	1
Energy Storage System		61.33	
Equipment Details	Storage cabinets 125kW/261kWh (Grid-forming)	59.1	12units
	Storage AC combiner panel 12×250A, 400VAC	2.23	1
Diesel Generator System		30.65	
Equipment Details	1000 kW DG set	20.9	1set
	ATS cabinet	3.75	1
	Synchronizing panel	3.0	1
	Container (noise/dust)	3.0	1
DVR System		25.95	

System	Main Investment Items	Estimated Amount (USD 10k)	Remarks
Equipment Details	DVR-800kW	3.5	1units
	STS 2000 kW	3.5	1
	Isolation transformer 0.4/0.4kV 800kVA	2.35	1
	Bypass switchgear ACB 2500A	3.75	1
	Battery cabinets 3×125 kWh	7.5	3
	DC combiner panel	1.2	1
	Container (DVR+control)	4.15	1
Microgrid Coordination Control System	Microgrid coordinator, gateway, GPS/Beidou clock, switch, server, workstation, O&M software, cabinet	15.0	1lot
Distribution System		17.56	
Equipment Details	SVG 300kVA	3.0	1units
	Dry-type transformer 1000 kVA 10/0.4 kV	3.0	1
	Utility incoming ACB 2000A -400V /AC	2.08	1
	Storage parallel incoming ACB 2500A -400V /AC	2.24	1
	PV parallel incoming ACB 2500A -800V /AC	2.24	1
	Load outgoing ACB 2000A -400V /AC	2.0	1

System	Main Investment Items	Estimated Amount (USD 10k)	Remarks
	Container (distribution)	3.0	1
Grid connection test, documentation, and training	1~2	5.5	1lot
Total equipment cost	-	216.19	
Gross margin	~10%	20	
Total Project Cost		236.19	
Remarks: Above prices exclude certification fees, on-site commissioning (remote program testing and guidance), on-site installation, external cables, and international shipping.			

10.2 Economic Benefits

- Annual PV generation: ~2,737,500 kWh.
- Annual electricity cost saving: Based on diesel generation cost ~\$0.26/kWh, saving ~711,750 US dollars per year
- Simple payback period: ~3.3 years (excluding financing costs). If the costs for installation, cables, shipping, etc. are included, it is expected to recover the investment within four years.
- Lifetime (10 years) total benefit: far exceeds initial investment
- Diesel reduction: ~3 million liters per year
- CO₂ reduction: ~8,000 tons per year

11. Project Implementation Plan

Phase	Work Content	Duration
Phase 1	Detailed design optimization: clarify requirements, custom equipment, specs	4-8 weeks
Phase 4	Manufacturing of storage, DG, PV, DVR systems	10-12 weeks
Phase 5	Electrical wiring and communication system commissioning	4-6 weeks
Phase 6	System integration, trial run, acceptance	2-4 weeks
Total	From start to operation	20-30 weeks

12. Risk Management and Mitigation

Risk Item	Impact	Mitigation Measures
Extreme heat (>50°C)	Battery performance degrades	Liquid cooling + shading design
Frequent sandstorms	PV output reduction, dust accumulation	Regular cleaning + sealed protection
Unstable diesel supply	DG unable to refuel	20,000 L fuel storage + alternative supply channels
Insufficient local O&M capability	Slow fault recovery	Remote monitoring + local training

13. Conclusion

This microgrid solution for the Sana'a factory adopts a four-in-one hybrid energy architecture: "PV + storage + diesel + storage-type DVR", achieving the following objectives:

- Uninterrupted power supply: Covers all five weather conditions, guaranteeing power for 10 working hours.
- Power quality: DVR ensures transfer time ≤ 5 ms and support ≥ 20 min during voltage dips/transients; SVG provides voltage compensation to keep sensitive load voltage fluctuation within $\pm 5\%$.
- Economic optimality: PV ≥ 1500 kWp + BESS ≥ 3132 kWh with best cost-performance ratio.
- Intelligent dispatch: Grid-forming PCS + EMS for fully automatic energy management.
- Safety and reliability: Multi-level protection, black-start capability, remote monitoring.
- Green and low-carbon: $\sim 8,000$ tons CO₂ reduction per year, reduced diesel dependence.

This proposal fully considers Yemen's local climate (low latitude, high irradiation, occasional sandstorms) and infrastructure conditions, using mature and reliable technology with modular design to ensure stable operation and economic returns throughout the microgrid's lifecycle.

Through deep AC coupling of PV and grid-forming storage, supplemented by diesel and storage-type DVR backup, this system achieves 10 hours of continuous high-quality power supply and seamless production under all weather conditions, meeting the plant's high-quality process requirements.

Project design:

Authorized units of Sunwoda : Guangzhou Jiancheng International Energy Co., Ltd. (Sunwoda Certificate No.:SUN-SES-001)

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